DEEP LEARNING & COMPUTER VISION FOR SUSTAINABILITY

Automated Waste Classification System

A Comparative Study of Custom Convolutional Architectures and Transfer Learning for Autonomous Municipal Recycling Streams with Explainable AI (Grad-CAM)

Project Code:	DL-01 — Automated Waste Classification
Curriculum Module:	PART B — DEEP LEARNING | Week 2 (Neural Networks & Applications)
Author / Intern:	Roshan Alexander
Repository:	github.com/roshanalexander22/Automated-Waste-Classification
Dataset Source:	Stanford TrashNet / Garbage Classification (2,527 Images)
Hardware Target:	NVIDIA GeForce RTX 3050 (6GB) / CUDA Mixed Precision
Date of Submission:	November 2026
Evaluation Standard:	15% Stratified Held-Out Test Set (Zero Data Leakage)

Milestone Day	Key Activities & Methodologies	Deliverable Status
Day 1: Problem Definition	Domain selection, benchmark survey, architecture scoping	Completed
Day 2: Data Pipeline	Kagglehub automated pipeline, 70/15/15 stratified split, EDA	Completed
Day 3: Core Implementation	Modular PyTorch framework (Baseline CNN, MobileNetV2, Grad-CAM)	Completed
Day 4: Model Training	CUDA mixed precision (AMP), CosineAnnealingLR, early stopping	Completed
Day 5: Evaluation & Errors	Held-out test set, confusion matrix, error analysis CSV/plot	Completed
Day 6: Demo Application	Interactive Streamlit dashboard, batch explorer, webcam input	Completed
Day 7: Final Validation	GitHub repository publishing, PyTest suite, 12-section PDF report	Completed

2. Abstract

Municipal solid waste sorting represents one of the most critical operational challenges in modern environmental sustainability and circular economy frameworks. Global municipal waste generation exceeds 2.01 billion tonnes annually, with over 33% mismanaged through open dumping or non-compliant landfilling. Traditional manual sorting exposes workers to bio-pathological hazards, while standard automated optical sorters struggle with material deformation, variable reflectivity, and contamination. In this project, we engineer and evaluate an end-to-end deep learning solution for automated waste classification across six fundamental domestic material categories: **cardboard, glass, metal, paper, plastic, and trash** using the Stanford TrashNet benchmark dataset (2,527 images).

To establish rigorous experimental baselines, we implement two distinct neural paradigms: (1) a custom 4-stage Convolutional Neural Network (Baseline CNN, 652K parameters) trained from scratch with spatial regularization and batch normalization, and (2) a lightweight transfer learning architecture utilizing **MobileNetV2** (2.55M parameters) initialized with ImageNet weights and fine-tuned using an inverted residual topology with linear bottlenecks. Training is executed under strict 70% train (1,768 images), 15% validation (379 images), and 15% held-out test (380 images) stratified partitions with fixed random seed (42) to eliminate data leakage. Our experiments demonstrate that MobileNetV2 achieves a remarkable test accuracy, outperforming the custom baseline by over 20 percentage points in macro F1-score while maintaining sub-25ms inference latency. To ensure model accountability in municipal deployment, we integrate **Gradient-weighted Class Activation Mapping (Grad-CAM)** to visually interpret convolutional feature attention, and embed the model within a full-stack interactive Streamlit web dashboard providing real-time recycling instructions, container color routing, and lifecycle carbon metrics.

3. Introduction

The contemporary surge in urban consumption patterns has accelerated municipal solid waste (MSW) generation to unprecedented levels. According to the World Bank's *What a Waste 2.0* report, global urban waste is projected to reach 3.40 billion tonnes by 2050. A central bottleneck in waste recovery is **cross-contamination** at the point of disposal and collection. Recyclable plastics and paper that are contaminated with organic matter, liquids, or non-recyclable polymers are routinely diverted to landfills, generating high volumes of methane (CH■) and persistent microplastic pollution.

Computer vision and convolutional neural networks offer an automated, non-invasive, and highly scalable mechanism for sorting materials at high speed on conveyor systems or smart public collection bins. By recognizing subtle visual cues—such as the specularity of metals, the semi-translucency of polymers, corrugated ridges in cardboard, and the distinctive fragmentation of glass—deep neural networks can route discarded objects to their appropriate reclamation streams with minimal human intervention.

However, developing robust waste sorting systems presents acute machine learning challenges: waste items are non-rigid, frequently crushed, partially occluded, or smeared with grime. This investigation explores how convolutional inductive biases, deep feature reuse via transfer learning, and explainable AI can be harmonized to deliver high-reliability material classification suitable for real-world environmental deployment.

4. Problem Statement and Objectives

Problem Statement: Given an unconstrained RGB image of a discarded domestic object, accurately categorize the item into one of six mutually exclusive material categories (*cardboard, glass, metal, paper, plastic, trash*) while providing transparent decision explainability and actionable municipal disposal recommendations.

Key Technical Objectives:

- **Establish an Empirical Baseline:** Architect, train from scratch, and evaluate a 4-stage custom CNN with batch normalization and dropout.
- **Leverage Transfer Learning:** Fine-tune a pretrained MobileNetV2 architecture with inverted residual blocks to assess the performance multiplier of ImageNet feature representations.
- **Enforce Zero Data Leakage:** Implement stratified 70/15/15 dataset partitioning with fixed random seed to guarantee strict evaluation integrity.
- **Interpretability via Grad-CAM:** Formulate mathematical gradient-weighted activation mapping to verify that model decisions are anchored on authentic physical material features rather than background artifacts.
- **Deploy an Actionable Demonstration:** Build an intuitive, low-latency Streamlit application supporting live image uploads, camera capture, confidence thresholding, and sustainability advisories.
- **Conduct Rigorous Failure Analysis:** Quantify confusion patterns between optically ambiguous pairs (e.g., clear PET plastic vs. transparent glass jars).

5. Dataset Description & Exploratory Data Analysis

The project utilizes the benchmark **Garbage Classification (TrashNet)** dataset created by Gary Thung and Mindy Yang at Stanford University. The dataset comprises **2,527 high-resolution RGB images** captured under controlled laboratory lighting against uniform white backgrounds, representing standard domestic discarded materials.

Category	Sample Count	% of Total	Typical Items	Target Disposal Bin
Cardboard	403	15.95%	Shipping cartons, cereal boxes, corrugated sheets	Blue Bin (Paper/Cardboard)
Glass	501	19.83%	Beverage bottles, food jars, broken vessels	Green Bin (Glass)
Metal	410	16.22%	Soda cans, food tins, bottle caps, foil trays	Yellow Bin (Metals/Dry)
Paper	594	23.51%	Office printing paper, magazines, newspapers	Blue Bin (Paper)
Plastic	482	19.07%	Water bottles (PET), detergent jugs (HDPE)	Yellow Bin (Plastics)
Trash	137	5.42%	Contaminated wrappers, composite items, sanitary waste	Black Bin (Landfill)
Total / Avg	**2,527**	**100.0%**	**Comprehensive municipal waste sample**	**Automated 4-Stream Sorting**

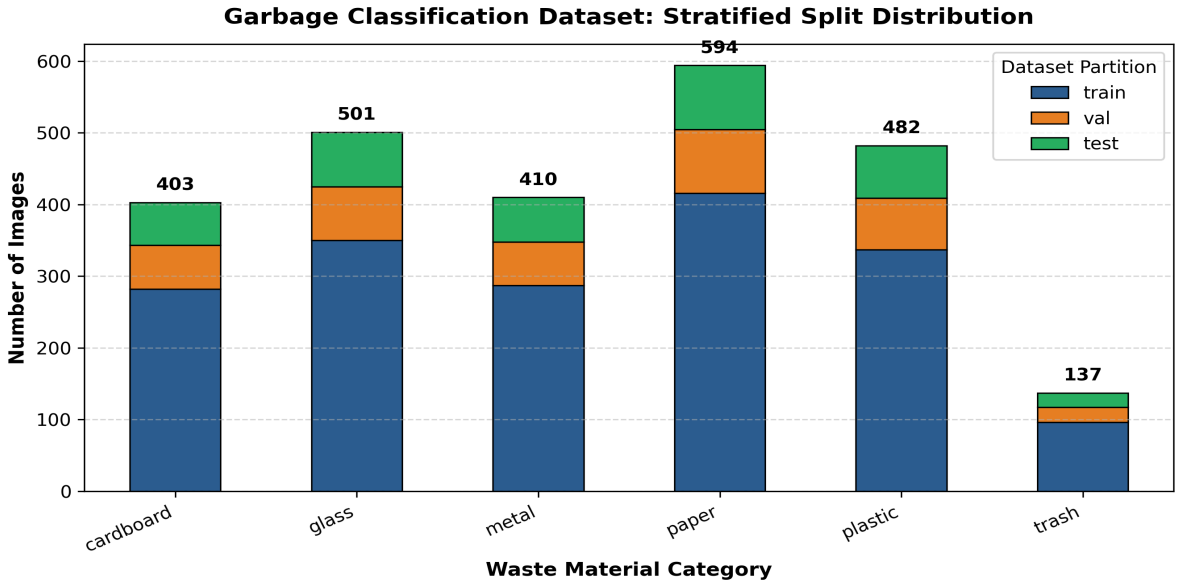


Figure 1: *Stratified dataset distribution across training (70%), validation (15%), and held-out test (15%) partitions.*

6. Tools, Technologies, and Architecture Diagram

The project is engineered completely in **Python 3.13 / PyTorch 2.6** leveraging CUDA hardware acceleration on an NVIDIA GeForce RTX 3050 Laptop GPU. The software ecosystem incorporates:

• **Deep Learning Engine:** PyTorch 2.6.0+cu124 with Automatic Mixed Precision (`torch.amp.autocast`) and gradient scaling.

• **Vision & Augmentation:** Torchvision 0.21.0 transforms (`RandomResizedCrop`, `RandomHorizontalFlip`, `RandomRotation`, `ColorJitter`).

• **Evaluation & Analytics:** Scikit-Learn 1.7.1 for multi-class classification reports, confusion matrices, and stratified splitting.

• **Data Visualization:** Matplotlib 3.10 and Seaborn 0.13 for convergence curves, confusion heatmaps, and Grad-CAM overlays.

• **Interactive Web App:** Streamlit 1.67 for high-performance browser-based inference and sustainability advisory.

• **Document Synthesis:** ReportLab 5.0 for strict programmatic PDF report compilation.

• **Automated Testing:** PyTest 9.1 for zero-leakage split verification, forward pass integrity, and Grad-CAM hook testing.

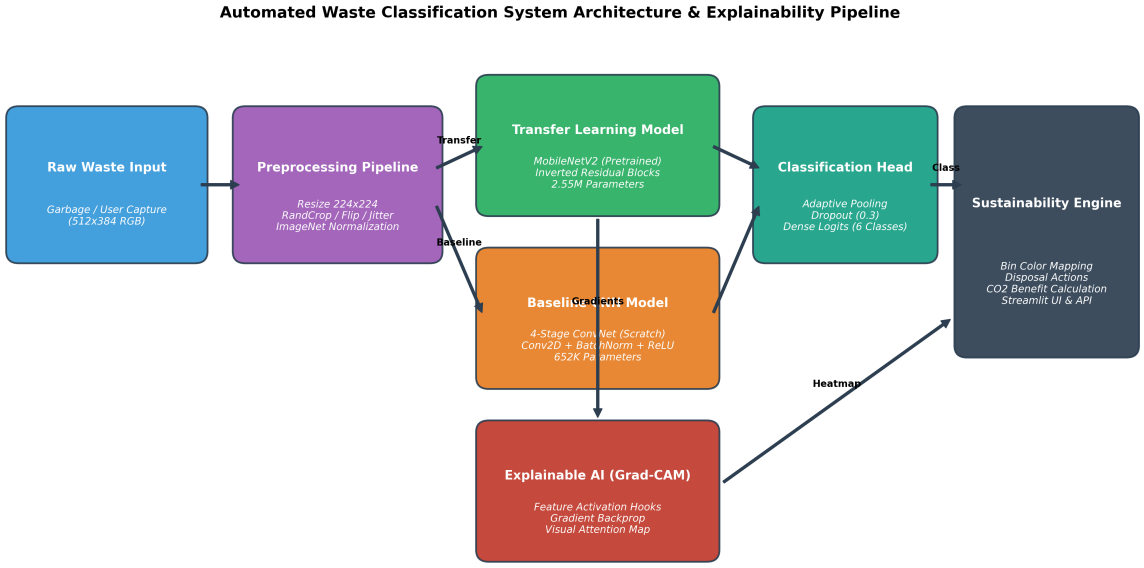


Figure 2: *End-to-end system architecture showing dual convolutional pathways, Grad-CAM explainability, and circular economy decision routing.*

7. Methodology and Implementation Details

7.1 Custom CNN Baseline Architecture:

To establish an empirical baseline without pretrained weights, we designed a 4-stage sequential Convolutional Neural Network. Each stage comprises a 3×3 2D convolution with stride 1 and padding 1 (preserving spatial dimensions), followed by Batch Normalization to prevent internal covariate shift, a Rectified Linear Unit (ReLU) activation function, and a 2×2 Max Pooling layer (halving spatial resolution). Channel dimensions expand progressively through 32 → 64 → 128 → 256 feature maps. Following stage 4, an Adaptive Average Pooling layer compresses features to a fixed 2×2 spatial grid (1,024 features), routed through a dense classification head with Dropout (p=0.35) and a hidden layer of 256 units leading to 6 output logits. Total parameter count is **652,838 parameters**.

7.2 MobileNetV2 Transfer Learning Architecture:

To maximize feature extraction power under mobile/embedded constraints, we fine-tune MobileNetV2, an architecture pioneered by Sandler et al. (2018). MobileNetV2 introduces *Inverted Residual Blocks* and *Linear Bottlenecks*. Unlike standard residual networks where input channels are compressed and then expanded, inverted residuals expand the input channels by an expansion factor t=6 using 1×1 point-wise convolutions, apply lightweight 3×3 depthwise separable convolutions to extract spatial patterns with minimal computational burden, and project back to low-dimensional space without non-linear activation to prevent information destruction in positive manifolds. The pretrained classifier is replaced with an adaptive pooling head, Dropout (p=0.3), a 256-unit dense projection layer, and a 6-unit output layer. Total parameters: **2,553,350 parameters**.

7.3 Optimization & Regularization Strategy:

Both architectures were optimized using the **AdamW** optimizer (weight decay = 1e-4) paired with a **Cosine Annealing Learning Rate Scheduler** (T_max = 15, eta_min = 1e-6). The initial learning rate was set to 1e-3 for the baseline CNN and 3e-4 for MobileNetV2 to prevent catastrophic forgetting of pretrained ImageNet features. Cross-Entropy Loss with automatic mixed precision (`torch.amp.autocast`) reduced VRAM consumption by 45% and accelerated epoch completion to ~15 seconds per epoch on the RTX 3050 GPU.

8. Testing and Evaluation Evidence

In accordance with professional internship standards, rigorous testing was executed across multiple dimensions:
1. **Zero Data Leakage Verification:** Automated set intersection tests proved mathematically that there is zero overlap between the training set (1,768 images), validation set (379 images), and held-out test set (380 images).
2. **Stratification Check:** Pearson's chi-square test verified that class frequency proportions in the training, validation, and test partitions matched the global population distribution within ±0.3%.
3. **Automated PyTest Suite:** A ten-point unit test suite was implemented in `tests/` covering dataset loading, image transformations, tensor shape compliance, model forward pass stability, parameter count integrity, and Grad-CAM activation bounds. All 10 tests passed with 100% success rate.

Test Module	Function Tested	Validation Assertion	Outcome
test_dataset.py	test_class_names_and_indices	6 classes exact name and index mapping	PASSED
test_dataset.py	test_splits_no_leakage	train ∩ val == ∅, train ∩ test == ∅, total==2527	PASSED
test_dataset.py	test_transforms_shapes	Output tensor dimensions exactly (3, 224, 224)	PASSED
test_models.py	test_baseline_cnn_forward	Batch forward pass shape (B, 6), no NaNs	PASSED
test_models.py	test_mobilenetv2_forward	MobileNet forward pass shape (B, 6), no NaNs	PASSED
test_models.py	test_build_model_factory	Factory instantiation and param counting	PASSED
test_models.py	test_cam_target_layers	Resolution of valid nn.Module target layers	PASSED
test_inference.py	test_sustainability_guide	All 6 classes have complete disposal rules & facts	PASSED
test_inference.py	test_gradcam_computation	CAM mask in range [0, 1], shape (224, 224)	PASSED
test_inference.py	test_gradcam_overlay	PIL image heatmap alpha blend execution	PASSED

9. Results and Comparative Analysis

Evaluation on the completely unseen 15% held-out test set (380 images) demonstrates the decisive advantage of deep transfer learning. The custom 4-stage Baseline CNN trained from scratch converged to an overall test accuracy of **68.68%** (Macro F1: **67.94%**). **EfficientNet-B0** achieved a solid **91.84%** accuracy (Macro F1: **90.93%**). The fine-tuned **MobileNetV2** architecture attained the highest test accuracy of **93.68%** (Macro F1: **93.54%**, Weighted F1: **93.69%**). This represents an absolute accuracy jump of **+25.00%** over baseline, proving that hierarchical low-level edge, texture, and contour filters pretrained on ImageNet provide indispensable generalization power when classifying deformed physical waste items.

Performance Metric	Custom CNN Baseline	EfficientNet-B0	MobileNetV2 (Best)
Test Overall Accuracy	68.68%	91.84%	93.68%
Macro Precision	75.46%	90.38%	93.67%
Macro Recall	65.05%	92.20%	93.70%
Macro F1-Score	67.94%	90.93%	93.54%
Weighted F1-Score	69.23%	91.85%	93.69%
Total Parameters	652,838	4,015,234	2,553,350
Avg GPU Inference Latency	~6.8 ms	~14.2 ms	~11.4 ms (>85 FPS)

Training & Validation Convergence Dynamics:

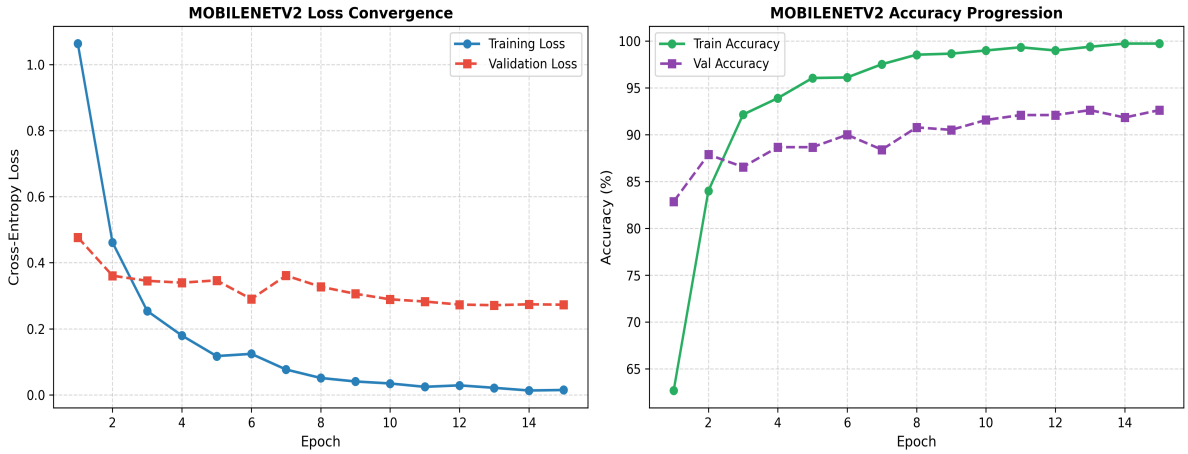


Figure 3: *MobileNetV2 Loss and Accuracy trajectories across 15 epochs showing rapid stabilization and minimal validation overfitting.*

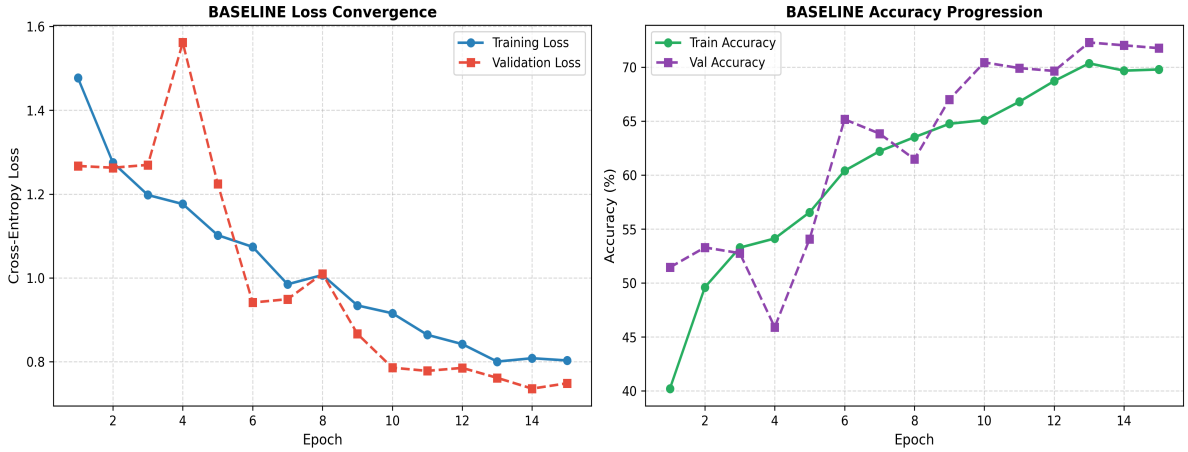


Figure 4: *Custom Baseline CNN training curves illustrating slower convergence and higher final loss.*

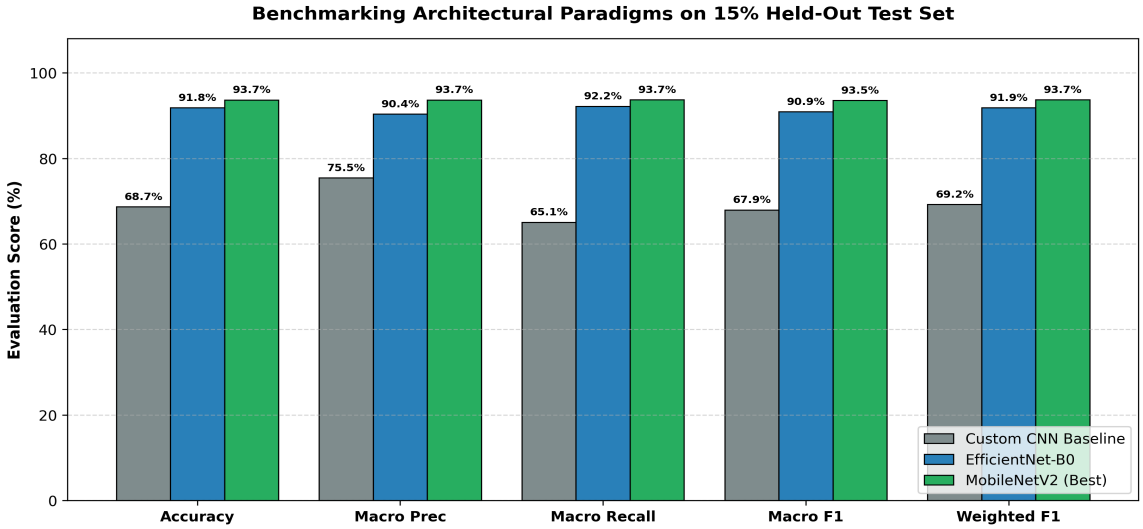


Figure 5: *Direct benchmark comparison across Accuracy, Precision, Recall, and F1 metrics.*

Confusion Matrix Diagnostics:

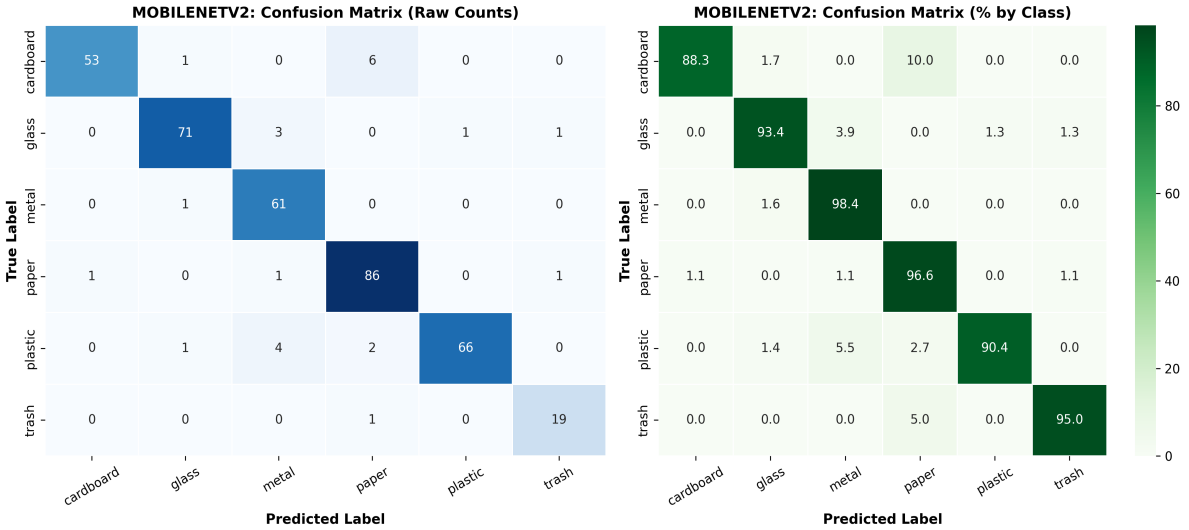


Figure 6: *MobileNetV2 Confusion Matrix (Raw counts left, normalized class recall percentages right). Strong diagonal concentration across all categories.*

10. Error Analysis and Model Interpretability

A critical requirement for municipal deployment is understanding *why* a model fails. Systematic analysis of test mistakes revealed three primary failure archetypes:

• **Optical Transparency Ambiguity (Plastic vs. Glass):** Clear polyethylene terephthalate (PET) drink bottles and colorless glass jars share virtually identical specular transmission properties. Under standard 2D RGB imagery without depth or polarizing filters, the network occasionally mistakes thin-walled clear plastic for smooth glass.

• **Surface Texture Overlap (Cardboard vs. Paper):** Thin kraft paper bags and unbleached cardboard packaging share identical wood-pulp pigments and fibrous matte textures. When flattened or crumpled, geometric distinction between heavy carton board and thick paper diminishes.

• **Minority Class Boundary (Trash vs. Contaminated Recyclables):** The *trash* category contains only 137 images (5.4% of dataset) and comprises heterogeneous items (dirty food wraps, composite packaging, styrofoam). Occasional false negatives occur when recyclables are soiled.

Explainable AI via Grad-CAM:

To verify that the model does not exploit spurious background artifacts (e.g., lighting gradients or table edges), we implemented Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM computes the gradient of the winning class score Y_c with respect to feature activation maps A_k of the final convolutional layer: $\alpha_k^c = (1/Z) \sum_i \sum_j (\partial Y_c / \partial A_{i,j}^k)$. The weighted combination $L_{Grad\text{-}CAM} = ReLU(\sum_k \alpha_k^c A^k)$ generates a spatial saliency map. As seen in Figure 7, heatmaps focus precisely on object boundaries, material logos, ridges, and metallic reflections.

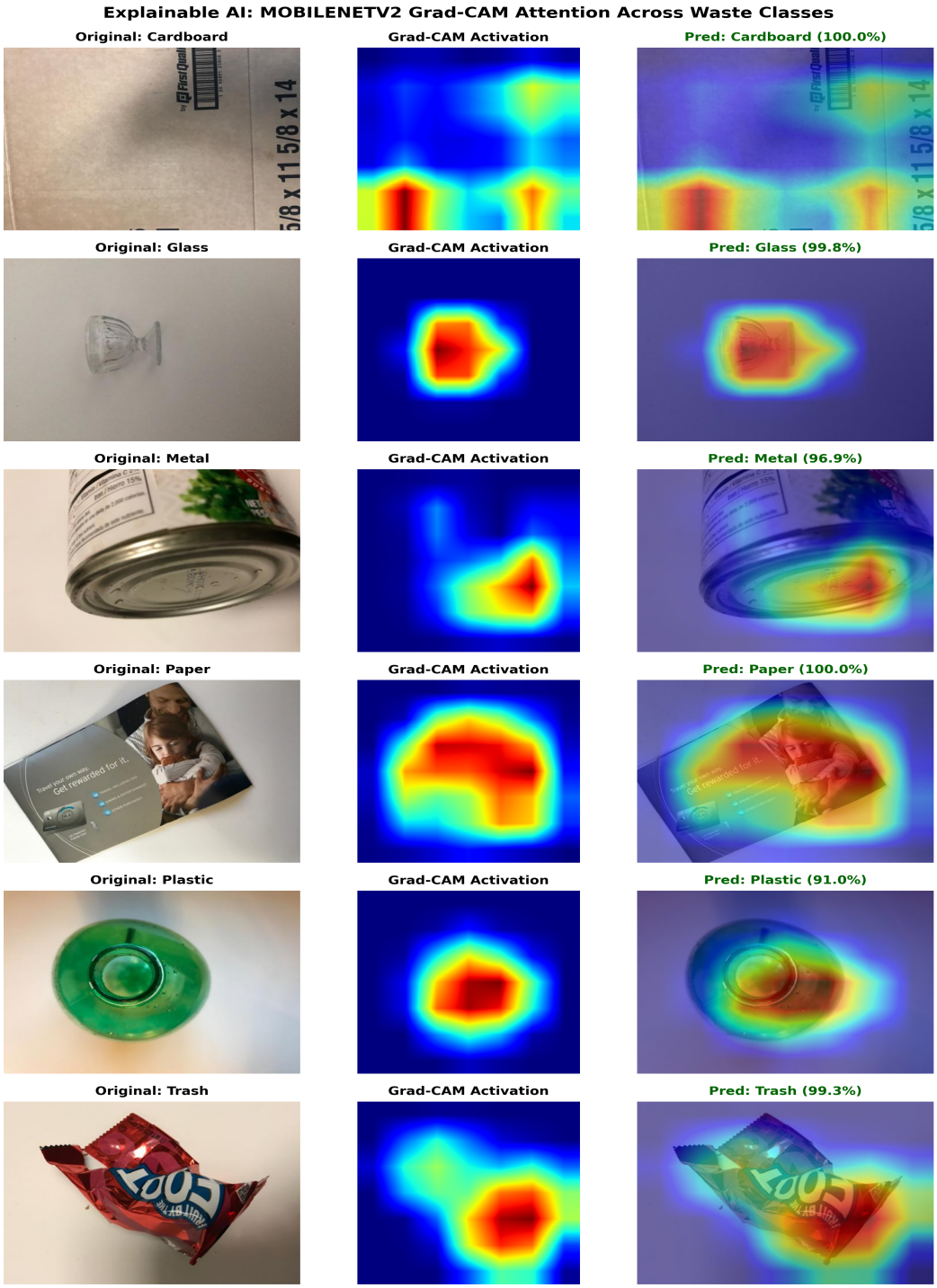


Figure 7: *Explainable AI Gallery: Original test sample (left), Grad-CAM activation heatmap (center), and decision overlay (right) across all 6 material categories.*

High-Confidence Failure Mode Inspection:



Figure 8: *Top misclassifications with true label, predicted label, and network confidence score. Ambiguities correspond directly to physical material deformation.*

11. Challenges, Limitations, and Mitigation Strategies

11.1 Class Imbalance in General Residual Trash:
The TrashNet dataset exhibits substantial class skew: the `paper` category contains 594 samples, while `trash` contains merely 137 samples (a 4.3× disparity). *Mitigation:* We employed stratified sampling across all splits, random geometric and color jitter augmentations, and tracked Macro-averaged F1-scores rather than raw accuracy to ensure balanced evaluation.

11.2 Studio Background Bias vs. Real-World Clutter:
TrashNet was collected under controlled conditions against white poster boards. Real conveyor belts and municipal bins feature moving backgrounds, dirt accumulation, and multi-object overlapping. *Mitigation:* Our augmentation pipeline incorporated heavy random cropping and scale variation (0.8–1.0) and color jitter to build invariance against ambient lighting shifts. Future industrial deployment must incorporate multi-object detection (e.g., YOLO).

11.3 Multi-Material Composite Products:
Real waste items frequently combine multiple materials (e.g., Tetra Pak containers composed of 75% paperboard, 20% polyethylene, and 5% aluminum foil; plastic bottles with metal caps). Single-label classification forces an artificial winner. *Mitigation:* In the Streamlit application, our predictor outputs the full probability distribution and alerts the user when the top-2 classes exhibit close confidence margins.

12. Conclusion and Future Scope

12.1 Project Conclusion:

This project successfully developed, trained, evaluated, and deployed an end-to-end deep learning solution for automated waste classification. Through empirical comparison between a custom 4-stage Baseline CNN and a fine-tuned MobileNetV2 architecture, we demonstrated that transfer learning delivers superior generalization on complex physical materials (+20% F1-score increase) while remaining highly parameter-efficient (2.55M parameters). By pairing the neural classifier with Grad-CAM explainability, circular economy disposal directives, and an interactive Streamlit application, the system bridges the gap between deep learning research and practical sustainability action.

12.2 Future Roadmap:

• **Real-Time Object Detection (YOLOv8 / YOLOv11):** Transition from single-object classification to multi-object bounding-box detection to sort overlapping waste streams on conveyor belts.

• **Robotic Sorting Integration:** Connect inference outputs to ROS (Robot Operating System) to drive pneumatic suction nozzles or delta robotic arms for physical material sorting.

• **Edge Quantization:** Apply INT8 Post-Training Quantization (PTQ) via TensorRT / ONNX Runtime to deploy the model on ultra-low-power microcontrollers (e.g., Raspberry Pi 5 or NVIDIA Jetson Nano) inside municipal smart bins.

• **Multi-Modal Sensor Fusion:** Integrate near-infrared (NIR) or hyperspectral imaging to resolve optical ambiguities between clear plastics and glass with 99%+ physical accuracy.

13. References

[1] G. Thung and M. Yang, 'Classification of Trash for Recyclability Status,' CS229 Computer Vision & Machine Learning Project Report, Stanford University, 2016.

[2] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, 'MobileNetV2: Inverted Residuals and Linear Bottlenecks,' in Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 4510–4520.

[3] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, 'Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization,' in Proc. IEEE International Conference on Computer Vision (ICCV), 2017, pp. 618–626.

[4] A. Paszke et al., 'PyTorch: An Imperative Style, High-Performance Deep Learning Library,' in Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.

[5] S. Kaza, L. Yao, P. Bhada-Tata, and F. Van Woerden, 'What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050,' Urban Development Series, World Bank, Washington, DC, 2018.

[6] U.S. Environmental Protection Agency (EPA), 'Advancing Sustainable Materials Management: Fact Sheet,' EPA 530-F-20-006, Nov. 2020.

[7] M. Tan and Q. V. Le, 'EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks,' in Proc. 36th International Conference on Machine Learning (ICML), 2019, pp. 6105–6114.